

SUPPORTING INFORMATION

Targeting the assembly of bacterial cell division protein FtsZ with small molecules

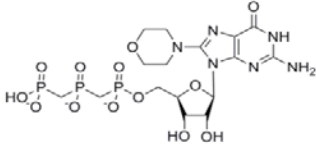
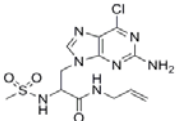
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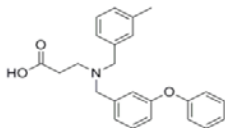
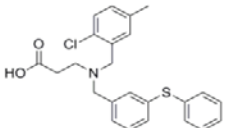
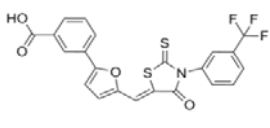
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SUPPLEMENTARY TABLE S1

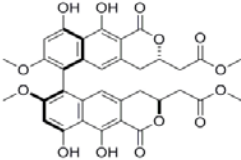
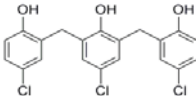
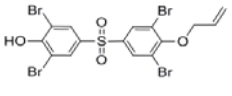
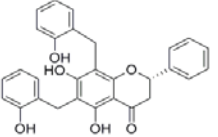
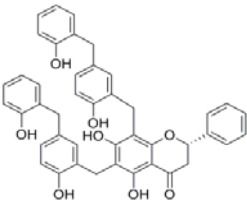
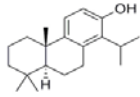
Guanine derivatives

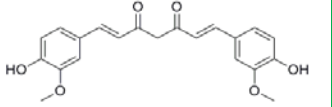
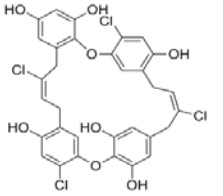
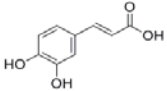
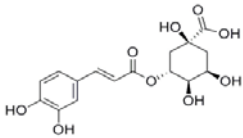
Name (references)	Structure MW (g·mol ⁻¹)	Effects on purified FtsZ			Binding site (Docking score in the GTP site) ^b	Effects on bacterial cell division / Z-ring	Antibacterial activity MIC (μM)	Cross-reactions with tubulin and eukaryotic cells
		Assembly ^a IC ₅₀ (μM)	GTPase IC ₅₀ (μM)	Other				
C8 GTP analogs: 8-MorpholinoGTP (1-3)	 601	110±15 (Ec, pel.), 139 (Ec, scatt.), 504±26 (Bs, pel.)	179±15 (Ec)	K _d = 0.43±0.01 μM (Mj, competition assay)	GTP site: crystal structure in complex with FtsZ determined. Competes with a fluorescent nucleotide (Medium)	nd	Ineffective (4)	C8-GTP analogs support microtubule assembly
Guanine derivatives: compound 14 (5)	 374	nd	600 (Pa)	nd	unknown	nd	Weakly inhibits growth of Sa ATCC 25923 at 53 mM	nd

Carboxylic Acids

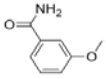
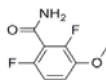
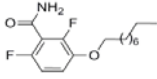
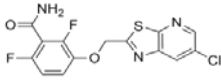
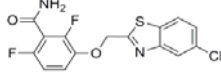
Name (references)	Structure MW (g·mol ⁻¹)	Effects on purified FtsZ			Binding site (Docking score in the GTP site) ^b	Effects on bacterial cell division / Z-ring	Antibacterial activity MIC (μM)	Cross-reactions with tubulin and eukaryotic cells
		Assembly ^a IC ₅₀ (μM)	GTPase IC ₅₀ (μM)	Other				
PC58538 (6)	 375	Inhibition (Bs, pel.)	362 (Bs)	nd	Resistant mutants in Bs: A26V, A47T, A49V and R184H map around the GTP site (Medium)	Filamentation in Bs at 341 μM / Disturbance of Z-ring at 341 μM (FP)	Weak. e.g. Bs, 341	nd
PC170942 (6)	 426	nd	~ 44 (Bs, competitive) 1100 (Ec)	nd	unknown	Filamentation in Bs at 19 μM	More potent than analog PC58538. e.g. Bs, 38	nd
OTBA (7)	 475	Increases assembly of Ec at 10 μM and Bs at 40 μM (scatt.) Bundling observed (EM)	40 (Bs)	K _d = 15.0±1.5 μM (Ec FtsZ-Y371W fluorescence)	unknown	Filamentation and disturbance of Z-ring in Bs at 1 μM (immuno.)	Bs, 2	No effect on tubulin assembly at 40 μM Inhibits HeLa cell proliferation (IC ₅₀ =8 μM)

Phenols and polyphenols

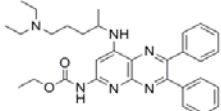
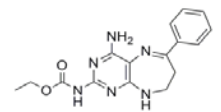
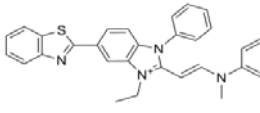
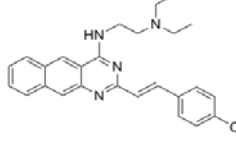
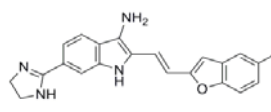
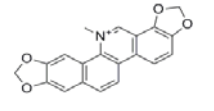
Name (references)	Structure MW (g·mol ⁻¹)	Effects on purified FtsZ			Binding site (Docking score in the GTP site) ^b	Effects on cell division / Z-ring	Antibacterial activity MIC (μM)	Cross-reactions with tubulin and eukaryotic cells
		Assembly ^a IC ₅₀ (μM)	GTPase IC ₅₀ (μM)	Other				
Viriditoxin (8, 9)	 663	12.3 (Ec FtsZ- T65C- fluorescein, filter.)	10.6 (Ec)	nd	unknown (High)	Filamentation in <i>sulA</i> deleted Ec at 19 μM	Gram + species MRSA, 12	Non-toxic to HeLa cells at 100 μM
Zantrin Z1 (10, 11)	 410	Inhibits at 10 μM (Ec, pel.) Supported by EM	5 (Ec) 55 (Mtb)	nd	unknown	No filamentation in Ec DRC39/ Partial effects on Z-rings at 10μM (immuno.)	Broad spectrum. More effective on Gram + species MRSA, ~ 3	nd
Zantrin Z4 (10)	 606	Inhibits at 50 μM (Ec, pel.) Supported by EM	25 (Ec) > 100 (Mtb)	nd	unknown	No filamentation in Ec DRC39	Gram + species MRSA, 10	nd
Dichamanetin (12)	 469	nd	12.5±0.5 (Ec)	nd	unknown	nd	Gram + species Bs, ~ 2 <i>S. aureus</i> , ~ 2	nd
2'''-Hydroxy-5''- benzylisouvarinol-B (12)	 681	nd	8.3±0.5 (Ec)	nd	unknown (High)	nd	Broad spectrum Bs, ~ 4 <i>S. aureus</i> , 11	nd
Totarol (13, 14)	 286	27% inhibition at 50 μM (Mtb, pel.) Aggregates in EM	~ 40 (Mtb)	K _d = 11.0±2.3 μM (ANS fluorescence)	unknown	Filamentation and disturbance of the Z- ring in Bs at 1.5 μM (immuno.)	Gram + species (15) Bs, 2	No detectable effects on microtubules in HeLa cells. HeLa cell proliferation IC ₅₀ =18±1 μM

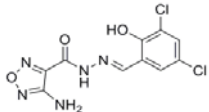
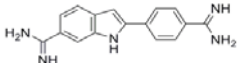
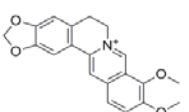
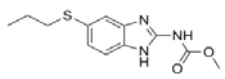
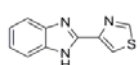
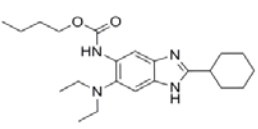
Curcumin (16, 17)	 368	30 (Ec, scatt.) Supported by EM	35% increase at 30 μ M (Ec)	$K_d = 7.3 \pm 1.8 \mu$ M (curcumin fluorescence)	Modelled into the GTP binding site (High)	Filamentation in Bs and disturbance of Z ring at 25/50 μ M (immuno.)	Bs, 100 Ec K12 MG16 $IC_{50} = 58 \pm 5 \mu$ M	Suppresses microtubule dynamics (18)
Chrysopaentin A (19)	 678	Inhibits at 50 μ M (Ec, EM)	9.9 ± 2.5 (Ec)	Binds to Ec FtsZ (STD-NMR)	GTP site: GTP γ S inhibits binding in STD-NMR, modelled into the GTP site	nd	MRSA MIC_{50} , 2 ± 1 Vancomycin resistant <i>Enterococcus</i> <i>faecium</i> MIC_{50} , 4 ± 1	Does not affect tubulin assembly up to 150 μ M Non-toxic to HCT-116 and P388 cells up to 74 μ M, and to BSC-1 cells up to 147 μ M
Caffeic acid (20)	 180	106 ± 6 (Ec, scatt.)	23% inhibition at 100 μ M (Ec)	nd	Docked to the T7 loop (Medium)	Filamentation in Bs at 100 μ M	Ec ATCC 25922, 1780 Bactericide at MIC (21)	nd
Chlorogenic acid (20)	 338	70 ± 4 (Ec, scatt.)	46% inhibition at 100 μ M (Ec)	Circular dichroism	Docked into the T7 loop (High)	Filamentation in Bs at 100 μ M	Weak inhibition of Ec. CM-871 at 500 μ g/well (22)	nd

Benzamide derivatives

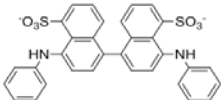
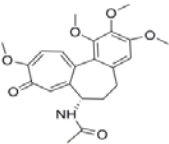
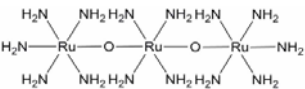
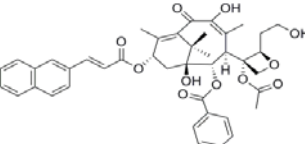
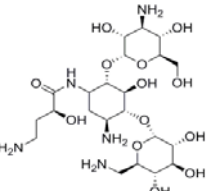
Name (references)	Structure MW (g·mol ⁻¹)	Effects on purified FtsZ			Binding site (Docking score in the GTP site) ^b	Effects on cell division / Z-ring	Antibacterial activity MIC (μM)	Cross-reactions with tubulin and eukaryotic cells
		Assembly ^a IC ₅₀ (μM)	GTPase IC ₅₀ (μM)	Other				
3-MBA (23)	 151	nd	nd	nd	unknown resistant mutants in Bs: A47P and V173F (Medium)	Filamentation in Bs UOT1285 in the mM range	Inhibits growth of Bs UOT1285 in the mM range	nd
2,6-Difluoro-3-methoxybenzamide (DFMBA) (24, 25)	 187	Induces assembly and condensation in the mM range (Bs and Sa, scatt.+pel.) Supported by EM	nd	nd	Interdomain cleft?	Filamentation in Bs at 1370 μM	Bs, 2740	No effect on microtubule assembly at 4000 μM
Alkoxybenzamides: compound 41 (24)	 299	nd	nd	nd	Modelled into the interdomain cleft	Filamentation of Bs at 0.41 μM and ballooning of Sa at 0.84 μM	Bs, < 1 Sa, ~ 2. Analogs produce resistant FtsZ mutants	Alkoxybenzamides have suboptimal ADME results (27)
PC190723 (25-28)	 356	Induces assembly and condensation in the μM range (Bs and Sa, scatt.+pel)	0.16 (Sa) Modulates activity in Bs and Ec	K _d ~ 10 μM to Bs polymers	Resistant mutants and molecular modelling at the interdomain cleft (High docking score in the cleft)	Filamentation of Bs and ballooning of Sa 5.6 μM / Disturbance of Z- ring at 5.6 μM (FP)	Bs, 2.8 MRSA, 2.8 Protects mice from a lethal dose of Sa ATCC 19636 ED ₅₀ =3.1 mg/kg	No effect on tubulin at 20 μM. HepG2 cell IC ₅₀ >180 μM ADME better than analog 8j
8j (27, 29)	 355	Induces assembly at μM range (Bs, scatt.+pel.) Curvature observed by EM	maximal inhibition at 56 μM (Bs, uncompetitive)	nd	Possibly the interdomain cleft	Filamentation of Bs at 0.6 μM. Helical division at 0.06 μM / Disturbance of the Bs Z-ring at 0.6 μM (FP)	Sa, ~ 1 Protects mice from a lethal dose of Sa Smith ED ₅₀ =41 mg/kg	Significant plasma protein binding and clearance from hepatocytes in ADME tests

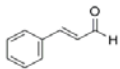
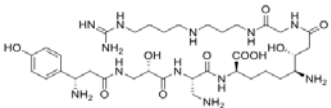
N-Heterocycles

Name (references)	Structure MW (g·mol ⁻¹)	Effects on purified FtsZ			Binding site (Docking score in the GTP site) ^b	Effects on cell division / Z-ring	Antibacterial activity MIC (μM)	Cross-reactions with tubulin and eukaryotic cells
		Assembly ^a IC ₅₀ (μM)	GTPase IC ₅₀ (μM)	Other				
SRI-3072 (30, 31)	 541	52±12 (Mtb, scatt.)	20% inhibition at 100 μM	nd	unknown	nd	Mtb MIC ₉₉ < 1 Bactericide MBC ~ 2 4 μM. Inhibits Mtb growth in macrophages	No effect on microtubule assembly at 100 μM. Vero cell IC ₅₀ = 12 μM
SRI-7614 (30)	 326	60 (Mtb, scatt.)	25% inhibition at 100 μM	nd	unknown	nd	Mtb MIC ₉₉ , 19	Microtubule assembly IC ₅₀ = 4 μM. Vero cell IC ₅₀ > 613 μM
Zantrin Z2 (10)	 488	Inhibits at 20 μM (Ec, pel.) Supported by EM.	10 (Ec) 70 (Mtb)	nd	unknown	No filamentation in Ec DRC39/ Partial effects on Z-rings at 5μM (immuno.)	Broad spectrum. More effective on Gram + species. MRSA, ~ 3	nd
Zantrin Z3 (10)	 431	Increases at 30 μM (Ec, pel.) Supported by EM.	15 (Ec) 50 (Mtb)	nd	unknown	No filamentation in Ec DRC39/ Inhibits Z-rings at 10μM (immuno.)	Broad spectrum, MRSA, 10	nd
Zantrin Z5 (10)	 410	Weakly increases at 10 μM (Ec, pel.) Supported by EM.	4 (Ec) 30 (Mtb)	nd	unknown	Filamentation in Ec DRC39 at 80 μM/ inhibits Z-rings at 40μM (immuno.)	Bs JH-642, 40	nd
Sanguinarine (32)	 332	Weak inhibition (Ec, scatt.+pel.) Supported by EM.	nd	K _d = 30±6 μM (FtsZ-ANS fluorescence)	unknown	Filamentation in Ec BL21 and Bs/ Disturbance of Z- ring at μM range (Bs- immuno and Ec-FP)	Bs IC ₅₀ = 3±1 Ec BL21 IC ₅₀ = 14±2μM	Inhibits microtubule assembly. HeLa cell proliferation IC ₅₀ = 1.6 μM (33)

4-Aminofurazan derivative A189 (34)	 316	40% reduction at 316 μ M (Ec, scatt.)	253 (Ec) Increasing with GTP concentration	nd	unknown, GTP site suggested by authors (Medium)	Filamentation and disturbance of Z-ring in Ec M101 at 203 μ M (immuno.)	Ec NIHJ JC-2, 405 Sa KMP9, 51	nd
DAPI (35)	 277	Delays depolymerization in the μ M range (Ec, scatt.). Bundling observed with EM	$K_i = 29.4 \pm 0.3 \mu$ M (Ec, uncompetitive)	$K_d \sim 17 \mu$ M (anisotropy)	unknown (Medium)	nd	nd	Interacts with tubulin (36)
Berberine (37, 38)	 336	10.0 $\pm$ 2.5 (Ec, scatt.) Supported by EM.	16 $\pm$ 5 (Ec)	$K_d \sim 0.02 \mu$ M (ITC) STD-NMR signals. Reduces bis-ANS-FtsZ fluorescence	Docked into the GTP site	Increases cell length and disturbs the Z-ring of Ec at 100 μ M (FP)	Ec K12, 1500 Genetic evidence of FtsZ targeting	nd
Albendazole (39)	 265	nd	nd	nd	unknown	Filamentation in Mtb and inhibition of septa formation at 60 μ M	Mtb, 60-120 Bactericide at 60 μ M	Disturbs microtubules in 1A9 cells. 1A9 cell $IC_{50} = 0.24 \mu$ M (40)
Thiabendazole (39, 41)	 201	nd	nd	nd	unknown	Filamentation in Mtb and inhibition of septa formation at 80 μ M	Mtb, 80-160 Bactericide at 80 μ M	Fungal tubulin inhibitor (42)
Trisubstituted benzimidazole 1aG7 (43)	 387	7.7 (Mtb, scatt.) Supported by EM.	Up to 400% enhancement at 80 μ M (Mtb)	nd	unknown	Cell shortening and altered morphology in Mtb at 4 μ M	Mtb, 2	Vero cell $IC_{50} > 400 \mu$ M

Others

Name (references)	Structure MW (g·mol ⁻¹)	Effects on purified FtsZ			Binding site (Docking score in the GTP site) ^b	Effects on cell division / Z-ring	Antibacterial activity MIC (μM)	Cross-reactions with tubulin and eukaryotic cells
		Assembly ^a IC ₅₀ (μM)	GTPase IC ₅₀ (μM)	Other				
bis-ANS (44)	 595	14 (Ec, scatt. + fluorescence microscopy)	nd	K _d = 1.33 μM and low affinity sites (fluorescent binding)	unknown (Medium)	nd	nd	Inhibits microtubule assembly in the μM range (45)
Colchicine (30)	 399	104±2 (Mtb, scatt.)	35% inhibition at 100 μM	nd	unknown	nd	nd	Inhibits tubulin assembly - IC ₅₀ = 6.5 μM. Known antimitotic agent.
Ruthenium red (46)	 560	Increases at 2 μM (Ec, scatt.+pel.) Bundles in EM	~ 2 (Ec)	Binds to polymers. Ca ²⁺ reduces binding.	unknown	nd	nd	Disturbs microtubules <i>in vivo</i> and is toxic in neurons at 100 μM (47)
Taxanes: compound 10a (48)	 725	nd	nd	nd	unknown	Filamentation in Mtb at 2.5 μM	Drug resistant Mtb IMCJ946.K2, ~1	MCF7 and A549 cell IC ₅₀ > 80 μM
Amikacin (49)	 586	nd	nd	nd	unknown (High)	Filamentation and Disturbance of Z-ring in Ec at 6.8 μM (FP)	Ec, lethal at 55 μM	nd

Cinnamaldehyde (50)		6.9±2.2 (Ec, scatt.) Supported by EM.	5.8±2.2 (Ec)	$K_d = 1.0 \pm 0.2 \mu\text{M}$ (ITC)	Docked into the T7 loop pocket. STD-NMR signals	Increases Ec cell length and disturbs Z-ring at 100 μM (FP)	Ec, 7580 Bs, 3790 MRSA, 1890 Bactericide at MBC = 7580 μM in Ec & Bs	Hemolysis not detected
Edeine B ₁ (51)		Inhibits at 20-160 μM (Ec, scatt.)	nd	FtsZ binds to an edeine B1 affinity column	unknown (Medium)	Filamentation in Bs at 30 μM / Disturbs Z-ring (FP)	Bs, 20 Bactericide, with DVC ₉₀ = 100 μM	nd

a- Note that compound concentrations higher than those used in 90° light scattering or pelleting assays are often employed in electron microscopy in order to better visualize effects.

b - Compound docking scores into the nucleotide site are indicated between brackets only when they are rated High or Medium. See numerical docking scores and rating criteria in Table S2.

Abbreviations employed for bacterial species: For effects on purified FtsZ, abbreviations refer to the species of origin of FtsZ. Bs, *B. subtilis*; Ec, *E. coli*; Sa, *S. aureus*; Mtb, *M. tuberculosis*; Mj, *M. jannaschii* and Pa, *P. aeruginosa*. For effects on bacterial cells: Bs, *B. subtilis* 168; Sa, *S. aureus* ATCC 29213; MRSA, Methicillin Resistant *S. aureus* and Mtb, *M. tuberculosis* H37Rv. The *E. coli* strain employed is cited if specified. When a different strain has been used, it is specified. Strains for FtsZ-fluorescent protein visualization *in vivo* are specific to each assay and are not cited.

Abbreviations of experimental procedures: ADME, Absorption, Distribution, Metabolism, and Excretion; DVC₅₀, Decreased Viability Concentration 50%; ED₅₀, Effective Dose 50%; EM, Electron Microscopy; Filter, Filtration assay; FP, Fluorescent Protein microscopy; IC₅₀, Inhibitory Concentration 50%; Immuno, Immunofluorescence microscopy; ITC, Isothermal Titration Calorimetry; K_d , Dissociation constant; K_i , Inhibition constant; MBC, Minimum Bactericidal Concentration; MIC, Minimum Inhibitory Concentration; nd, not done; Pel., Pelleting; Scatt., 90° Light Scattering and STD-NMR, Saturation Transfer Difference Nuclear Magnetic Resonance

In silico screening.

Given the lack of experimental data, *in silico* techniques offer an attractive alternative for assessing the binding ability of the compounds reviewed here. Two state of the art docking tools, ICM (52) and Glide (53), have been employed to give computational insights into the recognition of FtsZ sites by small molecules. Given the known limitations and the target dependence of scoring functions (54, 55) these methodologically different methods will cover a wider docking space. Standard procedures for flexible molecular docking have also been carried out with two different receptor 3D structures (Protein Data Bank IDs: 2VXY and 2VAP) to add more conformational variability.

Preparation of Receptor Structures and binding pocket definition.

Protein structures were regularized in accordance to the docking software used. In all cases, charges were assigned, orientations of side chain amides were corrected, and finally hydrogen were added and their positions optimized by energy minimization using the corresponding force field. To achieve a common definition of the binding pocket, the two receptor structures were first aligned. The boundaries of the binding box of nucleotide site were derived from the GTP-crystallized ligand coordinates of the 2VAP structure. The taxol-like pocket near key residues V307 (26) was localized using the ICM pocket finder utility.

Preparation of ligand structures

Compounds were compiled from the literature and 3D atomic coordinates, tautomeric forms, stereochemistry, hydrogen atoms, and protonation states were assigned using standard procedures. For docking with Glide, known ligand chirality was respected by Ligprep. Otherwise, at most 32 low energy input 3D stereoisomers per ligand were generated. The OPLS_2005 force field was chosen and the possible ionization states at $\text{pH} = 7.0 \pm 2.0$ were generated using Epik 1.6. In these structures generation of only the lowest ring conformation was considered. During the Glide docking procedures, flexible conformations of ligands were limited to variation on acyclic torsion bonds, pyramidal nitrogen inversions, and the ring conformations. Ligand molecules were prepared for the ICM docking by rotational search followed by a Cartesian minimization using MMFF force field in the absence of the receptor. These free molecules were optimized with global energy in the internal coordinate space and the lowest-energy conformations were used as starting points for ICM docking. During docking, the ligand torsional or roto-translational variables are randomly changed.

Docking Methods.

We employed standard protocols for flexible docking with the two methodologies. For Glide, the Extra-precision (XP) docking protocol was used for ligand docking followed by short energy minimization of the complex using the OPLS-AA (56) force field. The ICM methodology utilizes internal coordinates to optimize flexible ligands in a grid-based receptor field. Energies are computed using MMFF (57) partial charges with the ECEPP/3 (58) force field. The ligand docking was also complemented by local energy minimization using a Biased Probability Monte Carlo method (59). The docking scores obtained with the different receptor structures and methods are detailed in Table S2.

Supplementary Table S2

	NUCLEOTIDE SITE					INTERDOMAIN CLEFT SITE				
	ICM		GLIDE		Rating*	ICM		GLIDE		Rating*
	2VXY	2VAP	2VXY	2VAP		2VXY	2VAP	2VXY	2VAP	
GDP	-46.35	-50.36	-6.10	-6.40	High	-20.24	-2.78	-5.02	-6.27	Medium
GTP	-43.35	-57.37	-6.65	-8.05	High	-7.735	-7.553	-6.182	-5.22	Medium
8-MorpholinoGTP	-39.10	-43.84	-7.01	-2.08	Medium	-12.53	8.43	-5.59	-4.47	
Guanine derivative 14	-8.98	-21.60	-4.55	-3.70		-11.70	-15.33	-3.84	-5.54	
PC58538	-17.39	-20.3	-4.83	-5.23	Medium	-13.98	-6.11	-3.42	-5.50	
PC170942	-15.08	-16.86	-4.33	-3.28		-8.04	-6.27	-3.46	-5.63	
OTBA	-16.98	-18.19	-3.10	-3.40		-13.13	-6.21	-2.86	-4.22	
Viriditoxin	-20.44	-26.43	-6.32	-5.29	High	-13.78	-11.08	-4.46	-5.53	
Zantrin Z1	-19.86	-16.24	-4.54	-4.32		-10.82	-14.56	-3.58	-6.70	
Zantrin Z4	-14.76	-17.72	-4.63	-5.61		-13.31	-11.84	-3.67	-4.10	
Dichamanetin	-21.90	-11.27	-4.99	-4.92		-16.05	-8.54	-4.17	-5.68	
2'''-OH-5''-benzylisouvarinol-B	-24.42	-22.64	-6.68	-6.95	High	-14.95	-17.69	-6.57	-7.71	Medium
Totarol	-4.26	-19.29	-3.41	-5.05		-7.21	-3.65	-3.57	-4.00	
Curcumin	-29.18	-36.07	-5.45	-8.19	High	-9.36	-12.98	-5.43	-5.53	Medium
Chrysopaentin A	-12.28	-13.55	-5.20	-4.07		-6.18	-8.41	-3.19	-4.17	
Caffeic Acid	-31.05	-30.11	-4.97	-4.00	Medium	-25.05	-10.17	-3.94	-5.70	
Chlorogenic acid	-36.00	-31.79	-7.44	-6.68	High	-26.09	-19.06	-6.57	-7.62	Medium
3-MBA	-20.18	-20.73	-4.01	-4.20	Medium	-22.24	-16.60	-3.70	-4.12	
DFMBA	-15.31	-19.49	-3.95	-3.91		-17.65	-9.38	-3.73	-4.10	
Alkoxybenzamide C41	-13.21	-22.11	-2.45	-3.80		-12.98	-3.92	-3.44	-3.43	
PC190723	-16.42	-23.39	-4.32	-4.32		-27.84	-23.94	-5.87	-6.05	High
8j	-17.69	-20.23	-4.51	-4.46		-26.87	-12.78	-3.82	-3.18	
SRI-3072	-4.77	-12.49	-5.08	-4.11		-12.45	2.78	-4.41	-6.20	
SRI-7614	-14.82	-7.88	-2.79	-0.56		-11.14	-3.13	-1.26	-3.17	
Zantrin Z2	-3.89	-13.79	-3.83	-2.81		-8.43	-16.26	-2.39	-4.28	
Zantrin Z3	-0.03	-12.32	-2.16	-3.22		-8.98	-14.52	-4.38	-5.01	
Zantrin Z5	-8.53	-23.09	-2.63	-2.63		-7.26	-14.10	-2.92	-3.79	
Sanguinarine	-12.95	-15.90	-3.71	-2.29		-8.63	-3.59	-2.52	-4.46	
4-Aminofurazan derivative A189	-22.54	-23.12	-4.08	-3.13	Medium	-17.76	-11.84	-3.48	-3.00	
DAPI	-29.60	-31.00	-4.80	-3.56	Medium	-17.51	-30.21	-4.74	-5.12	Medium
Berberine	-5.96	-16.56	-4.06	-3.08		-13.36	-5.13	-3.91	-4.42	
Albendazole	-10.48	-19.55	-3.14	-4.63		-14.45	-6.81	-3.85	-4.21	
Thiabendazole	-9.74	-13.87	-3.08	-1.63		-13.94	-5.52	-3.20	-3.86	
Trisubs. benzimidazole 1aG7	-8.31	-8.44	-2.93	-2.35		-15.33	-6.60	-3.23	-6.80	
Bis-ANS	-31.27	-24.17	1.90	-3.93	Medium	-11.01	5.32	-2.41	-4.58	
Colchicine	-13.89	-17.28	-3.24	-3.07		-11.81	-9.05	-3.39	-3.56	
Taxane derivative 10a	-16.26	-14.72	-5.06	-4.76		-2.03	-10.64	-2.78	-5.14	
Amikacin	-25.89	-27.53	-10.43	-8.05	High	-2.83	-1.90	-8.05	-9.48	Medium
Cinnamaldehyde	-11.99	-13.21	-3.08	-3.12		-16.72	-18.49	-2.78	-3.49	
Edeine B ₁	-21.95	-53.14	-2.97	-3.28	Medium	-23.59	-10.65	-6.20	-7.40	Medium

*Overall docking rating: A compound is defined to have a high docking score if positive dockings were obtained with the two docking methods (ICM, Glide) and with the two crystallographic structures (2VXY, 2VAP). The docking is positive when the docking score is smaller than -20 or -5 for ICM and Glide, respectively. A medium score is used when only one of the docking methods had positive scores with the two receptor structures or a receptor structure had positive scores with both docking methods

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